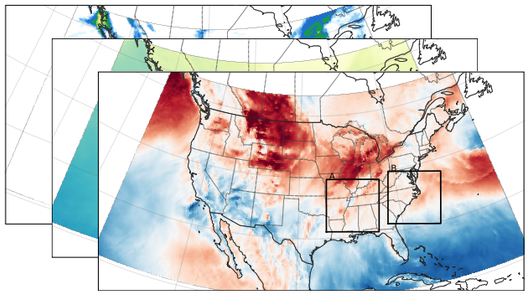
# MERRA21C-ML: GENERATIVE CONUS DOWNSCALING

GEOS-FP (~25 km) → Conditional flow matching → GEOS grid (~3 km)

## A COARSE INPUTS

GEOS-FP · ~25 km



**Weather at t − 1 h and t**
Rain rate · PRECTOT
10 m winds · U10M, V10M
Column vapor · TQV
2 m humidity · QV2M
Sea-level pressure · SLP

**Coarse baselines at t**
T2M, PRECTOT, U10M, V10M
Surface pressure · PS

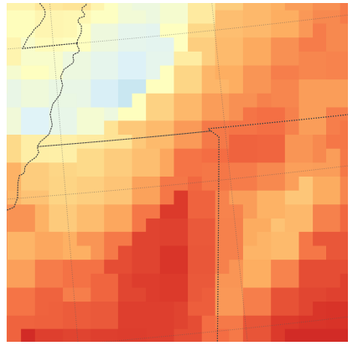
**Static fields**
Elevation · lat/lon · cell area
Land/lake/ocean fractions
Water distance · x/y slopes
Lake-availability mask

**Time (sine/cosine)**
Annual · UTC · local solar

## B RESOLUTION

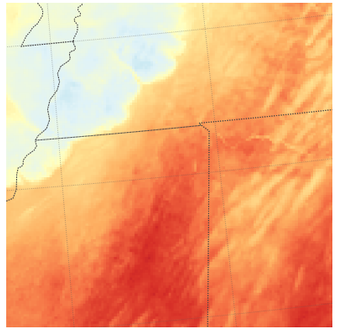
Same extent · same scale

**GEOS-FP · ~25 km**






**GEOS replay · ~3 km**

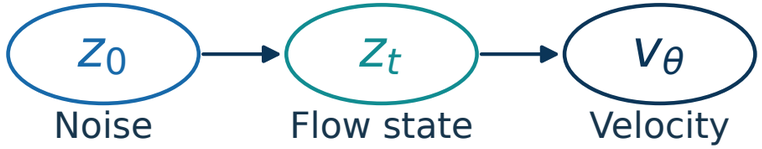





**Temperature (°C) · 3× zoom**

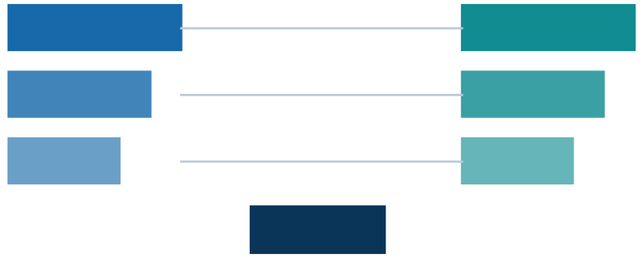
## C FLOW MATCHING + ENSEMBLE GENERATION

Frozen regression · local patches + broad context




$$z_t = (1 - t)z_0 + tz_1$$

**Conditional U-Net**

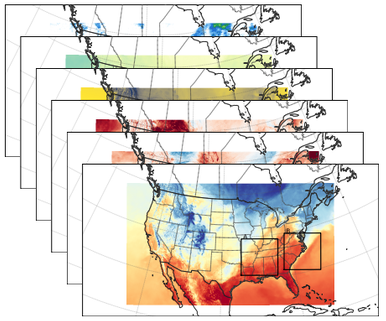


**Weighted velocity loss**



Core + halo
Area + channels

**Six output fields**



**~3 km / member**

$T_2m$ · rain · $p_s$
$u_{10}$ · $v_{10}$ · $q_2m$

**8 members**
Independent seeds

**Tiled Heun → decode → ensemble**

Blend tile velocities; advance one shared domain state